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Session : Technical Paper Presentation

Agentic AI for India's Utilities: From Loss Reduction to Grid-Edge Orchestration

Presented By

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Introduction

India's Power Sector is at an Inflection Point

Context

- India's power sector is undergoing **rapid digital and clean-energy transition**, driven by RDSS, smart metering rollout, and rising DER penetration
- National AT&C losses have shown improvement but remain **15–16% on average**, impacting utility financials
- **Smart metering rollout is accelerating**, but implementation remains uneven across states. This limits full utilization of AMI data for operations, billing, and analytics
- Rising penetration of **rooftop solar, BESS, and EVs** is reshaping the grid edge, increasing the need for real-time orchestration of distributed assets
- Utilities are also balancing digital modernization with **growing IT/OT cybersecurity risks** due to expanded connected infrastructure

Why Now?

- DISCOMs are expected to support **India's 500 GW RE target**, electrification, and emerging grid-edge models
- High data availability (AMI, SCADA, GIS, DER), but **low data-to-decision conversion**
- Need for intelligent, autonomous, auditable decision frameworks that reduce losses, accelerate outage restoration, and automate DER/VGI coordination

India stands at a pivotal moment—modern infrastructure is expanding, but systemic challenges continue to constrain efficiency, reliability, and financial sustainability

Challenges

High AT&C Losses

Persistent AT&C losses reduce financial viability and slow modernization investments in utilities.

Fragmented IT/OT Systems

Siloed core systems hinder real-time awareness and comprehensive analytics in utility operations.

Grid Edge Visibility

Rapid rooftop solar, EVs, and storage adoption outpaces orchestration frameworks, limiting control.

Cybersecurity and Compliance

Growing digitalization increases cybersecurity risks and complicates compliance with regulations.

Agentic AI can shift DISCOMs from reactive operations to intelligent, proactive grid orchestration

How Agentic AI can solve Utilities Challenges

Building on this shift toward intelligent and proactive operations, Agentic AI delivers practical, domain-aligned solutions across utility functions

Autonomous Agent Operations

Agentic AI deploys autonomous agents operating within defined objectives and constraints to monitor utility systems

Loss Reduction and Outage Management

Agents correlate utility data to detect losses and automate outage triage for faster fault resolution

DER and Vehicle-Grid Integration

Agentic AI coordinates solar, battery storage, and EV charging to support grid stability and peak shaving

Human-in-the-Loop Controls

Agentic AI operates with human oversight, audit trails, and escalation for transparency and safety

Key AI Agents Proposed for Utility Transformation

Loss Hunter Agent

Detects non-technical losses by analyzing consumption anomalies and historical billing data to improve revenue assurance

Outage Triage Agent

Automates fault analysis and FLISR workflows reducing outage duration and enhancing customer notifications

DER & Vehicle-Grid Orchestrator

Manages distributed energy resources and EV loads to ensure grid stability and resource adequacy compliance

Revenue Assurance Agent

Predicts payment risks and optimizes collection strategies to improve financial health of utility companies

CyberSentinel Agent

Continuously monitors IT and OT environments to detect cyber threats and ensure compliance with security guidelines

Adoption Lifecycle for Agentic AI in Utilities

Foundation Readiness Phase

Focuses on data availability, API enablement, integration architecture, safety envelopes, and human oversight mechanisms.

Pilot Deployment Phase

Targets high-impact areas with pilot deployments using KPIs like AT&C reduction and outage management improvements.

Scaling and Integration Phase

Scales agents across zones, integrates into workflows, and introduces advanced orchestration like DER coordination.

Full Autonomous Orchestration

Multiple AI agents collaborate real-time across grid operations, commercial, and cybersecurity, with MLOps monitoring.

Responsible AI and Governance Framework

Risk-Based AI Classification
The governance framework uses risk-based classification to manage low, medium, and high-impact AI use cases effectively.

Data Governance and Compliance
Data governance aligns with privacy laws focusing on consent, minimization, and secure handling of consumer information.

Technical Guardrails for Safety
Human-in-the-loop, audit trails, explainability, and rollback mechanisms maintain transparency and accountability in AI actions.

Regulatory and Operational Alignment
AI deployments comply with grid codes and cybersecurity guidelines to ensure safe and ethical utility operations.

USE CASE- DER & Vehicle-Grid Orchestrator

At 6:30 PM, rooftop solar drops sharply

EV charging demand begins to surge (offices closing, fleets returning to depots)

Feeder-level load forecasts show two DTs crossing the 80% thermal limit.

The Agent detects the rising stress and orchestrates:

- Smart delay of 30–45 minutes for non-urgent EV charging
- Discharge command to community BESS (where available)
- Curtailment signal to select flexible DERs (policy-permitted zones)
- Notification to control center for critical feeder oversight

Grid remains stable **without outages**, **without transformer overloading**, and within **resource adequacy compliance**

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THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

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